

The Pure Truth About Latin *S Impurum*: Exposing the Fiction of Extrasyllabicity

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The analysis of so-called “*s impurum*,” treated in Steriade (1982, 1988), Jacobs (1992), Lehmann (2005), and Cser (2012, 2016), highlights the complexities and challenges involved in understanding Latin syllabification. *S impurum* refers most basically to word-initial *s*-plus-stop clusters, such as in *stā.re* ‘to stand’, *spērō* ‘I hope’, and *scrī.ba* ‘scribe’, corresponding to Spanish *es.tar*, *es.pe.ro*, *es.cri.ba*. This Latin onset cluster can also occur between two consonants: *dexter* [dekster] ‘right hand’. The term *impurum* carries a strong pejorative bias, defined as “unclean, filthy, foul” (Lewis and Short 1879). This label is perhaps due to a few properties:

- (1) Among the class of obstruent consonants, the continuant [s] is generally taken to be more sonorant than a stop consonant, yet of the two, it stands further from the syllable’s nucleus. In terms of sonority, this arrangement does not conform to the sequencing patterned elsewhere in Latin, and it is dispreferred cross-linguistically (though obviously not in English).
- (2) Within a word, *s impurum* can typically be integrated into the preceding syllable as a coda, as observed in *res.tā.re* ‘to stay behind, remain,’ derived from *stā.re* ‘to stand,’ prefixed with *re-*, and in underived *ves.per* ‘evening’. This syllabification of *s impurum* is well-supported by metrical evidence from poetic scansion (Lehmann 2005: 29) and is not considered controversial.
- (3) The heterosyllabicity of medial *s impurum* clusters (Cser 2012), e.g. *res.tā.re*, is perceived to be inconsistent with respect to the homosyllabicity of other clusters that are also prefixed: stops preceding L include *dē.+cli.nō*, ‘I bend’, *dē.+fla.grō* ‘I burn down’, *dē.+gluttiō* ‘I swallow down’; preceding R, *dē.+gre.dī* ‘to go/march down’, *re.+frāc.ta* ‘broken up’, *re.+tra.hō* ‘I bring back’.¹
- (4) An analysis of Latin poets in Cser (2012) shows a conspicuous avoidance of word-initial *s impurum* immediately following a short vowel in a given line—this is a telling pattern, as this particular “non-neutral” position is unique in exposing, through meter, the poet’s deliberate choice regarding which syllable the *s* should belong to. Each of the two possible ways to syllabify *s impurum* in this position results in the preceding syllable being scanned as heavy in one case, light in the other. By avoiding this context altogether, poets managed to sidestep a situation where the metrical effect of their preference for

¹ A cursory search of online Latin dictionaries did not reveal the prefixes *dē-* or *re-* preceding stems in *br*, *bl*, *dr*, although they may exist. The cluster *dl* constitutes a legitimate gap in the stop-liquid inventory.

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syllabifying the *s* would otherwise be made readily apparent. Presumably, these poets were reluctant to commit to one specific syllabification over the other.

The syllabification of *s*-plus-stop therefore appears to present sufficient curiosities to lead researchers to consider Latin clusters like *st*, *sp*, and *sc* in onset position to be problematic, in line with the subjective connotations of “*impurum*.” However, it is important to reevaluate the implications of these properties and whether the “impure” label and its connotations of unsightliness are truly warranted.

To account for these facts, previous researchers have argued that *s impurum* is “extrasyllabic” (Jacobs 1992, Cser 2012) proposing that the segment has no affiliation with any syllable and is simply unparsed. This use of the term appears to have evolved from the concept of “stray adjunction” (Steriade 1982, 1988). These devices that have been conjured up to explain away apparent “sonority reversals” are, for a frank Parker (2002:9), “painfully ad hoc.” Indeed. Crucially, Cser’s analysis depends on a positional restriction on extrasyllabic segments prohibiting them from neighboring a syllable nucleus, thus necessitating “leftward resyllabification.” This process incorporates most cases (but see Section 6) of word-medial *s impurum* into the preceding syllable as its coda, thus “explaining” the syllable division of words like *res.tā.re* and *ves.per*. However, as we’ve seen, the extrasyllabicity analysis relies on significant assumptions and interconnected stipulations to support its central claim.

In contrast, the proposed analysis necessitates that all segments be syllabified on the surface, with these syllabifications being optimal for their respective contexts. By employing an Optimality Theoretic (OT) approach, the reliance on concepts such as “extrasyllabicity,” “stray adjunction,” positional stipulations, and “resyllabification” will be eliminated. This alternative analysis centers on a hierarchy of ranked and violable constraints that are language-agnostic and cross-linguistically supported. These constraints provide valuable phonological insights and drive the entire analysis within a parsimonious and efficient OT solution. As a result, all segments are syllabified in a single pass without exceptions, eliminating the need for ad hoc resyllabification. It will soon become evident that resyllabification is not permissible within the Optimality Theory framework.

A testament to the robustness and parsimony of this analysis, Section 3 effortlessly accounts for other clusters that exhibit similar, superficially “inconsistent” syllabification patterns, as well as the seemingly exceptional effect of *x* on metrical structure. The OT perspective also sheds light on the poets’ avoidance of *s impurum* in “non-neutral” positions (Cser 2012, 2016), and—when poets chose not to shy away from it, though this was rare—their particular preferences for syllabifying it as an onset versus coda will be depicted in Section 5 in terms of the optimal output of a constraint hierarchy featuring an additional constraint.

Furthermore, through the lens of the proposed constraint hierarchy, we will examine in Section 6 the purportedly extrasyllabic “trapped” [s], e.g. *dexter* [dekster] ‘right’, whose syllabification Cser deems unknowable.

The structure of this paper may appear unconventional, as I’ve chosen to present my proposed Optimality Theory (OT) analysis before summarizing and critiquing the previous generative analysis in Cser (2012, 2016). This

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deliberate sequencing enables a more efficient comparison of the strengths and advantages of the proposed OT analysis when reviewing Cser's work. The sequence of topics will follow this outline:

1. Key data for Optimality-Theoretic analysis

2. Optimality-Theoretic analysis

- 2.1 A brief introduction to Optimality Theory
- 2.2 Tips for interpreting Optimality tableaux
- 2.3 A minimally viable constraint inventory
- 2.4 The tableaux
- 2.5 Occam's Razor

3. Applications to superficially unrelated phenomena

- 3.1 Other clusters: *ps*, *mn*
- 3.2 The ambisyllabicity of *x*

4. Debunking "extrasyllabicity"

- 4.1 Previous analysis (Cser 2012, 2016)
- 4.2 Critique

5. Demystifying "non-neutral" *s impurum*

- 5.1 "Neutral" vs. "non-neutral" position
- 5.2 An Optimality-Theoretic understanding of non-neutral *s impurum*
- 5.3 The significance of non-neutral *s impurum* & the poets' avoidance

6. C[s]C: Setting "trapped" [s] free

7. Conclusion

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1. Key data for Optimality-Theoretic analysis

This is simply an overview of the patterns to be accounted for. Optimality Tableaux 2.4.1–2.4.4 will demonstrate the phonological well-formedness of a select set of Latin’s basic syllable configurations, laying the groundwork for the more specific cases of *s impurum* addressed in Tableaux 2.4.5–2.4.6. These in turn will be repurposed as templates for populating subsequent tableaux.

TABLE 1: Canonical patterns

Tableau	Written	Input	Output, *ill-formed	Gloss
2.4.1	SUPRĀ	/su.prā/	[su.prā] *[sup.rā]	‘above’, prep., adv., prep.
2.4.2	ABRUPTA	/ab + rupta/	[ab.rup.ta] *[a.brup.ta]	‘broken off’, e.g. neu.pl.
2.4.3	REFRĀCTA	/re + frag + ta/	[re.frāk.ta] ² *[ref.rāk.ta]	‘broken up’, e.g. neu.pl.
2.4.4	ABĪRE	/ab + īre/	[a.bī.re] *[ab.ī.re]	‘to go away’
2.4.5	RESTĀRE	/re + stāre/	[res.tā.re] *[re.stā.re]	‘to remain’, ‘stay or be left behind’,
2.4.6	STĀRE	/stāre/	[stā.re] *[es.tā.re]	‘to stand’

² For the phonetic transcriptions of short and long mid vowels, the International Phonetic Alphabet will be put aside in favor of short [e], [o] and long [ē], [ō], thus circumventing the controversy over the relative tongue height of these vowels. The traditional dichotomy consists of short [e], [o] versus long [e:], [o:], [+ATR], long vowels being higher than their short counterparts.

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2. Optimality-Theoretic Analysis

2.1 A brief introduction to Optimality Theory

Since entering circulation in the early 1990s, Optimality Theory (OT) has offered a refreshing perspective on linguistic patterns by proposing that universal constraints interact with language-specific rankings to determine the optimal output based on a given input.

For readers new to OT, it may be helpful to shift from understanding phonology as “ $X \rightarrow Y$ ” (“ X becomes Y ”) derivations to a *performance-based competition* among top contenders, or “candidates,” vying for the title of “winner” or “sole survivor,” metaphors for the phonological optimality of the surface realization. The selection process is graphically represented in tableaux, where all of the candidates are evaluated in series against one constraint at a time, and the candidate that emerges as the sole survivor is indicated as optimal. It “wins” as the surface representation.

In OT, constraints are violable, sharply contrasting with the rigidity and lack of “forgivingness” of traditional generative rules. “Violability” in OT means that a candidate can remain in the competition even if it violates certain constraints, as long as it satisfies higher-ranked constraints better than competing candidates. Constraint ranking is thus critical to the selection process. While the constraints are considered to be universal, cross-linguistic variation results from the variety of possible input forms and the language-specific rankings of the myriad possible constraints. We will see how the notion of ranked and violable constraints brings fresh insight to the comparison of Latin words like *stā.re* with Iberian *es.tar*.

By conceptualizing phonological well-formedness as the result of constraint interaction, OT has contributed significantly to the field of phonological analysis. Its groundbreaking approach provides a powerful and insightful alternative to traditional rule-based, derivational frameworks, as will soon be demonstrated.

2.2 Tips for interpreting Optimality tableaux

The purpose of the Optimality tableau is to demonstrate which potential output form is worthy to be the optimal surface realization and which are suboptimal, along with the reasons for rejection.

Each row in the tableau represents a candidate for the optimal surface realization of the input, labeled (a), (b), and so on. These candidates are evaluated against a series of constraints, appearing in **BOLD SMALL CAPS** and listed in columns from left to right, as shown in Tableau 2.2.

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Tableau 2.2: Example

/input/, [output]

Descriptive blurb goes here

/input/	CONSTRAINT #1	CONSTRAINT #2	CONSTRAINT #3
☞ (a) Candidate A			**
(b) Candidate B		*!	*
(c) Candidate C	*!		

Constraints are ranked left to right by their importance, with higher-ranked constraints taking precedence over lower-ranked ones. The evaluation process begins with *all* candidates being first assessed against the top-ranking constraint, CONSTRAINT #1. When the performance of all candidates has been evaluated on this first constraint, all are then evaluated according to the second constraint, and so on. The serial nature of the evaluation of candidates ensures that all are examined for compliance with a particular constraint before being allowed to proceed to the next.

The optimal candidate is the one that best satisfies the highest-ranked constraints, *even if it violates lower-ranked ones*.

This scenario is exemplified in the above tableau. Candidate (c)'s immediate violation of the first constraint leaves candidates (a) and (b) still in the running. Candidate (b) fails on the second constraint, making (a) the winner *by default*, even though (a) twice violates the lowest-ranked (rightmost) constraint. As (a) has been established as optimal as of the second constraint, its two violations of the lowest constraint are considered “minimal” and irrelevant, as optimality has already been determined.

For clarity, let's review some basic notational conventions used in Optimality Theory (OT)...

- A constraint violation is marked with an asterisk (*).
- A fatal violation of a constraint, which eliminates the candidate from the competition, is indicated by a grayed-out cell containing “*!”, the exclamation point indicating fatality.
- An empty box (□) indicates the site of an inserted (epenthetic, paragogic, euphonic) segment whose existence is hypothetical or whose specifications are irrelevant, e.g. English *cat*[s], *dog*[z], *hors*[□z].
- A broken line dividing two constraint columns, instead of the usual solid line, means that it is unknown or irrelevant which constraint dominates which. A solid line means the one to the left dominates the one to the right.

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- Once a candidate has been declared eliminated (or optimal), any cells that are up for further evaluation are shaded gray, indicating that the candidate’s performance against these lower-ranked constraints has no bearing on the outcome at this point. Nonetheless, they will continue to be evaluated in the tableau for several reasons: (1) to determine or refine ranking for related scenarios in the study, (2) for comparative analysis of marginal differences that might highlight overlooked patterns, and (3) exhaustivity, as blank cells could leave the reader with an incomplete understanding of the phenomenon under scrutiny.
- This optimal candidate is marked with a pointing finger (), indicating its victor status.
- The “much greater than” sign ($\gg$) denotes the domination of one constraint over another; “C1 $\gg$ C2” means “C1 outranks (or ‘dominates’) C2.”

2.3 A minimally viable constraint inventory

The cast includes the following, in order of appearance...

ONSET	Every syllable must have an onset. This constraint favors CV over V.
No CODA	A syllable cannot contain a coda segment. This constraint favors V over VC.
L-ALIGN (M, σ)	The left edge of a morpheme must align with the left edge of a syllable.
SON SEQ	The Sonority Sequencing Principle. Within an onset cluster, sonority increases towards the nucleus, and decreases within the coda away from the nucleus. Inversely stated, onset clusters decrease in Consonantal Strength (CS), while coda clusters increase in CS.
MAX-I/O	Material in the input must be realized in the output. Basically: “Don’t delete”!
DEP-I/O	Material in the output must depend on, or correspond to, material in the input. This constraint penalizes feature insertions, structure-filling, and epenthesis.
L-ALIGN (LxWd, PRWD)	The left edge of the lexical word (LxWd) must align with that of the prosodic word (PrWd). This constraint penalizes, e.g., word-initial epenthesis in * $\square$ stār.
O-CONTIG	The order of segments in the output must correspond to the order of segments in the input. This constraint penalizes any “skips” or insertions,” as these would disrupt the input’s contiguity.

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L-ALIGN (LxWd, σ)

The left edge of the lexical word (LxWd) must align with that of a syllable (σ).

2.4 The tableaux

We'll begin by demonstrating the optimality of a few syllable structures basic to Latin to set the stage. Tableau 2.4.1 illustrates the optimality of the syllabification *su.prā*, as opposed to ill-formed **sup.rā*.

Tableau 2.4.1: SUPRĀ 'above'

/suprā/, [su.prā]

Stop + liquid ("muta cum liquida") onset clusters

/suprā/	ONSET	NO CODA
☞ (a) su.prā		
(b) sup.rā		*! [p.]

Neither candidate fails on the higher-ranking constraint since both syllables start with a consonant. Moving on to the second constraint, candidate (b) is delivered a fatal blow for incurring a violation, giving (a) the victory.

Although this tableau does not itself require the dominance of ONSET over NO CODA to arrive at the optimality of *su.prā*, the logical necessity of this ranking will soon become evident over the course of the ensuing tableaux.

The syllabification of stop + liquid ("muta cum liquida") deserves special attention, as it depends on whether or not a morphological boundary intervenes. If one does, the cluster is heterosyllabic, as demonstrated by *ab.+rup.ta* in Tableau 2.4.2, while in the absence of such a boundary, the cluster is homosyllabic, as in *re.+frāc.ta*, Tableau 2.4.3.

Tableau 2.4.2: ABRUPTA 'broken off'

/ab+rupta/, [ab.rup.ta]

Heterosyllabic stop + liquid cluster

/ab+rupta/	L-ALIGN (M, σ)	NO CODA
☞ (a) ab.+rup.ta		* [b], *[p]
(b) a.b+rup.ta	*! [.b+r]	* [p]

Tableau 2.4.2 introduces the constraint L-ALIGN (M, σ), which requires that the left edge of a morpheme (M) align

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with that of a syllable (σ), demonstrating the need for Alignment to outrank NO CODA. If this ranking were reversed, the optimal form would be candidate (b), **a.brup.ta*.³

Candidate (a)'s two NO CODA violations are inconsequential to its victory, since competitor (b) has already been defeated on the higher-ranking alignment constraint.

With no need for modification, the same constraint hierarchy parsimoniously accounts for the contrasting homosyllabicity of the stop + liquid cluster in, e.g. *re.frāk.ta* 'broken up', which will now be demonstrated in Tableau 2.4.3.

Tableau 2.4.3: REFRĀCTA 'broken up'

/re+frag+ta/, [re.frāk.ta]

Homosyllabic stop + liquid cluster

/re+frag+ta/	L-ALIGN (M, σ)	NO CODA
☞ (a) re.+frāk.+ta		* [k.]
(b) re+f.rāk.+ta	*! [+f.r]	** [f.], [k.]

³ Interestingly, this is a point of contention in Brazilian Portuguese pedagogy. Some Brazilian educators continue to recommend the pedantic pronunciation *ab-rup-to* [a.bi.'xu.pi.to], which showcases morpheme-syllable alignment. In contrast, the colloquial reality, *a-brup-to* [a.'bru.pi.to] disregards the word's etymology. A similar inconsistency surrounds the pronunciation of *sublinhar* 'to underline', realized colloquially as [su.blĩ.'jah] but carefully as [su.bi.lĩ.'jah], the latter making the morpheme boundary prosodically salient. It would be interesting to explore the historical roots of this tension surrounding morphological alignment in Romance-language prescriptivism.

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The alignment constraint in this tableau⁴ maintains syllable boundaries, similar to its role in Tableau 2.4.2, *ab.+rup.ta*. Alignment prevents the [f] of the stem from serving as a coda, leading to the elimination of candidate (b), **ref.rac.ta*, due to its misaligned juncture. Consequently, candidate (a), which satisfies the higher-ranking alignment constraint, emerges as the winner.

The parsimony of this solution lies in the ability of the alignment constraint to distinguish between syllable boundaries in different contexts. When comparing the optimality of *re.frāc.ta* with that of *ab.rup.ta* in their respective tableaux, we see that the same constraint, Alignment, accounts for both cases. In the former, the optimal cluster is homosyllabic, syllabified as an onset, while in the latter, it is heterosyllabic. This versatility of the alignment constraint demonstrates its elegance and efficiency in evaluating the well-formedness of competing syllabifications, asserting its value in affording parsimonious solutions.

Tableau 2.4.4 next illustrates the necessity for ONSET to dominate L-ALIGN (M, σ), which has just been established to dominate NO CODA.

Tableau 2.4.4: ABĪRE ‘to go away’

/ab+īre/, [a.bī.re]

“Exceptional” misalignment of morpheme and syllable boundaries

/ab+īre/	ONSET	L-ALIGN (M, σ)	NO CODA
(a) ab.+ī.re	! [.i]		* [b.]
☞ (b) a.b+ī.re		* [.b+ī]	

While Tableau 2.4.3 demonstrated that L-ALIGN (M, σ) must dominate NO CODA, Tableau 2.4.4 shows that

⁴ Omitted from this tableau are candidates and constraints unrelated to the *s impurum* cluster, e.g., candidates **[re.frag.ta]*, **[re.frak.ta]*, or constraints governing regressive voicing assimilation ([...gta], [...gda]). The long vowel in *-frāc-* reflects a regular form-meaning correspondence in Latin verbs (Kuryłowicz 1968:296), seen in the alternation between the short vowel in the **present** tense *lēgit* ‘reads’ (active), *lēgitur* (passive) versus a long vowel in the **perfect** *lēxit* (active), *lēctum* (passive). Conversely, the vowel in the **present** tense verbs *combūrit* (active), *combūritur* (passive) is long, while a short vowel identifies the **perfect** tense in the verbs *combūssit* (active), *combūstum* (passive). In OT terms, this pattern results from a high-ranking “correspondence” or “identity” constraint—I propose IDENT(PERF-wt)—ensuring that the stem syllable/vowel show consistent agreement in weight/length across all perfective forms, the passive following the lead of the active; hence, **perfective** active ~ passive alternations such as *lēgit* ~ *lēctum*, *refrēgit* ~ *refrāctum*; *combūssit* ~ *combūstum*, *pōsuī* ~ *pōsitum*. IDENT(PERF-wt) executes on the greater principle of Economy, as the generalization itself leverages the redundant and predictable nature of perfective forms—vowel length need only be specified once, in the perfect active, and IDENT(PERF-wt) obliges the perfect passive to comply.

An apparent contradiction arises in the perfective system of *cadere* ‘to fall’: *cecidī* (active) ~ *cāsum* (passive). The disagreement in length can be understood as the consequence of compensatory lengthening: *cād+um* > *cāssum* > *cāsum*.

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ONSET must dominate L-ALIGN (M, σ). This ranking prioritizes the need for syllables to have onsets over the alignment of morphological boundaries with syllable boundaries. In other words, this tableau tells us that Latin values syllables with onsets more than syllables and morphemes that are aligned. These insights reveal aspects of Latin's phonological character—an aspect we will see in striking contrast with Spanish shortly. The extrasyllabicity analysis overlooks and neglects these insights, hindering a comprehensive understanding of Latin's phonology.

Cser (2012) specifies the *a.bī.re* pattern as an exception, set against the presumed primacy of *ab+rupta* [ab.rup.ta] and *re+frācta* [re.frāk.ta]: “Resyllabification cannot override the syllable structure that is created within morphological boundaries except if onsetless syllables would result following a closed syllable (as in *a.bī.re*)” (p. 366).

In contrast, there is no exceptionality on the OT account, as the optimality of *a.bī.re* emerges through the natural consequence of the constraint ranking ONSET $\gg$ L-ALIGN (M, σ). This ranking is integrated into Latin's overarching constraint hierarchy and captures a necessary generalization about Latin phonology, thus challenging the characterization of the *a.bī.re* pattern as exceptional.

Now that we have laid the groundwork for Latin syllabification, we can now address the two syllabifications of *s impurum*, beginning with medial *s impurum* first, as in *restāre*, Tableau 2.4.5, followed by the cluster in initial position, as in *stāre*, Tableau 2.4.6. The syllabification of word-initial *s impurum* in connected speech and poetry is a more nuanced topic and will be considered in Section 5.

Tableau 2.4.5: RESTĀRE ‘to remain’

/re+stāre/, [res.tā.re]

“*S impurum*,” medial position

/restāre/	SON SEQ	L-ALIGN (M, σ)	NO CODA
(a) re.stā.re	*! [.st]		
☞ (b) res.tā.re		* [+s.t]	* [s.]

Candidate (a) aligns the morpheme and syllable at the left edge, satisfying the alignment constraint. However, this alone is insufficient to satisfy the Sonority Sequencing Principle, which in this case requires that sonority progressively increase through the onset cluster toward the nucleus. Owing to its high ranking in Latin, this principle plays a crucial role in determining the sequencing of consonants within Latin consonant clusters.

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Sonority is reckoned according to the following scale.

(5) Sonority scale, based on Burquest & Payne (1993)

Vowels > Glides > Liquids > Nasals > Fricatives > Affricates > Stops⁵

The inverse is the scale of Consonantal Strength (CS), according to which the Sonority Sequencing constraint can be alternatively stated: Onset clusters decrease in CS toward the nucleus; codas clusters increase in CS.

Above, candidate (a) **re.stā.re*, is immediately disqualified due to its violation of the relatively high-ranking SON SEQ constraint in Latin. In this syllabification, the slope of sonority decreases as we move from [s] to [t]. This violation proves fatal, making (b) the winner.

Due to the serial nature of the evaluation process, it does not matter that the optimal candidate (b) *res.tā.re* has incurred more violations than the eliminated candidate (a) **re.stā.re*. Optimality has already been decided by the top-ranking constraint, SON SEQ, so any constraints ranked lower have no impact.

The above Tableau 2.4.5 offers a valuable insight into the phonological nature of Latin, revealing that compliance with the SON SEQ constraint takes precedence over alignment. This preference explains why *s impurum* clusters are regularly misaligned in forms like *res.tā.re*, *obs.tā.re*, *sus.pi.ce.re*, and *pros.pec.tus*.

The analysis has thus far revealed a hierarchy of constraints governing some basics of Latin syllabification. In perhaps an unconventional order, we have just demonstrated the optimality of the derived form *res.tā.re* without having first dealt with its base form, *stā.re*, our canonical example of *s impurum*. The optimality of this form is now depicted in Tableau 2.4.6.

⁵ Within their respective categories, voiced fricatives, affricates, and stops are considered less sonorant (or “stronger” consonants) than their voiceless counterparts.

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Tableau 2.4.6: STĀRE ‘to stand’

/stāre/, [stā.re]

“*S impurum*,” word-initial

/stāre/	MAX-I/O	DEP-I/O	SON SEQ	ONSET	L-ALIGN (LxWD, PRWD)	NO CODA
☞ (a) stā.re			* [.st]			
(b) □s.tā.re		*! >□<		* [.e]	* [e.+s]	*[s.]
(c) s□.tā.re		*! >□<				
(d) tā.re	*! <s>					

(As a reminder, the white box (□), as seen in candidates (b) and (c), marks the site of an inserted segment whose existence is hypothetical or whose specification is irrelevant.)

A few additional constraints must be considered to understand the optimality of fully syllabified words like *stāre*, *spēs*, and *scrība*. MAX-I/O and DEP-I/O are universally fundamental, preserving the “faithfulness” of the input to the output and vice versa:

- MAX-I/O ensures that the input finds *maximal* expression in the output, thereby preventing any deletions from occurring. Informally stated: “Don’t delete.” Segments violating MAX-I/O (deletions) are indicated by angled brackets pointing outward, e.g., <s>.
- DEP-I/O ensures that every segment in the output is *dependent* on a corresponding segment in the input, thus preventing insertions (“epenthesis”). Informally: “Don’t fill.” Offenders of DEP-I/O (insertions) are indicated by angled brackets pointed inwards, e.g., >e<, epenthetic >□<.

It has been determined in a comprehensive survey of Latin nasal coda behaviors (Giangola, forthcoming) that MAX-I/O must outrank DEP-I/O in order for the constraint hierarchy to correctly select [sūmpta] for SŪMPTA assuming input /sum+ta/⁶ ‘having been taken (neu.nom./acc.pl. or fem.nom.sg.)’ and not *[sūn.ta], despite the temptation of the latter’s attractive homorganic nasal-stop cluster.

⁶ The grammar will supply the vowel length via IDENT(PERF-wt), which ensures the same mora count across all perfect tenses, based on the conceptually more basic voice, the active. Although the details of this constraint will be discussed more thoroughly in a later footnote, it is important to note that this is the reason passive /sumta/ appears without a macron.

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The high ranking of these faithfulness constraints eliminate candidates (b), (c), and (d), which were chosen as candidates for this tableau as they represent common “repair strategies” employed by speakers of various languages when confronted with word-initial *s*-plus-stop onset clusters basic to languages like English.

In committing an insertion violation, candidates (b) and (c) each commit a “lesser included offense,” to use a term from criminal law. Candidate (c), **se.tā.re*, inserts a vowel to satisfy SON SEQ, but in doing so, it also violates the constraint O-CONTIG (from “Output Contiguity,” McCarthy & Prince 1995), which penalizes insertions specifically within a morpheme. Meanwhile, candidate (b), **es.tā.re*, also inserts a vowel to satisfy SON SEQ, thus incurring a concomitant violation of L-ALIGN (LXWD, PRWD), which requires that the left edge of the lexical and prosodic word boundaries be in alignment—this penalizes insertions at the left edge. (O-CONTIG could have been included as well in the preceding tableau to eliminate (c) **s□.ta.re* for Latin STĀRE, with no effect on the optimal outcome.)

As a point of interest, and in passing, we are now situated in a position to understand the well-formedness of *s impurum* descendants like *es.tar* ‘to be’ in the Iberian languages (e.g. Portuguese, Galician, Castilian, Aragonese, Catalan) as the fallout of the constraint ranking: SON SEQ » MAX-I/O » DEP-I/O » O-CONTIG » ONSET, L-ALIGN (LXWD, PRWD), NO CODA. In other words, to satisfy inviolable SON SEQ in Spanish, deleting is worse than inserting, and inserting medially is worse than initially, even though the word-initial insertion creates a syllable without an onset, with a coda, and with misaligned lexical-prosodic boundaries. This optimality of Iberian *es.tar* is thus depicted in Tableau 2.4.7.

Tableau 2.4.7: Iberian ESTAR ‘is’ (condition, location)

/sta/, [es.ˈta]

“*S impurum*,” word-initial

/star/	SON SEQ	MAX-I/O	DEP-I/O	O-CONTIG	ONSET	L-ALIGN (LxWD, PRWD)	NO CODA
(a) star	*! [.st]						* [r]
☞ (b) □s.tar			*! >□<		* [t]	* [+V.sC]	* [s], [r]
(c) s□.tar			*! >□<	* s□...	* [t]		* [r]
(d) tar		*! <s>			* [t]		* [r]

A more specific set of markedness constraints fills the epenthetic vowel with the features of [e]. That said, this example offers a novel understanding of the difference between Latin and the Iberian languages in terms of a distinct prioritization of related phonological preferences. The chief distinction between the way Spanish and Latin each handle *s impurum* is that Spanish has promoted SON SEQ to a ranking now higher than that of DEP-I/O, the reverse of Latin’s ranking. Spanish further ensures that the DEP-I/O violation (epenthesis) occurs outside the bounds of the lexical word by penalizing insertions made within the word, the effect of O-CONTIG.

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Iberian *es.tar* is also an effective example of the importance of constraint ranking and violability, given the numerous low-ranking infractions incurred by optimal candidate (b).

2.4 Occam's Razor

It is somewhat ironic that Cser invokes Occam's Razor, which favors simpler explanations over more complex ones, when arguing for the extrasyllabicity of *s* in *stā.re* with respect to its coda status in *res.tā.re*. As demonstrated in the previous tableaux, the inherently exceptional construct of extrasyllabicity proves unnecessary, calling into question its adherence to this heuristic principle.

Unencumbered by extrasyllabicity, all surface realizations can now be understood as emerging naturally, epiphenomenal of the following ranking schema, or “dominance chain”:

(6) Constraint ranking

MAX-I/O » DEP-I/O » SON SEQ » ONSET » L-ALIGN (M, σ) » No CODA

This ranking schema effectively captures the surface realizations without resorting to the exceptionality of extrasyllabicity, showcasing the parsimonious nature of this approach and its adherence to Occam's Razor. In contrast to the extrasyllabicity analysis, this OT analysis accurately accounts for the observed patterns on the surface without ever needing to refer to [s] or to introduce stipulations. Instead, the analysis leverages independently-motivated and necessary phonological generalizations to drive a comprehensive and coherent explanation, demonstrating its robustness and parsimony.

3. Applications to superficially unrelated phenomena

Having demonstrated the efficacy of this OT analysis in resolving the apparent contradiction in the syllabification of *s impurum*, we now turn our attention to additional phonological “inconsistencies” whose explanation emerges naturally from the proposed analysis, without modification or stipulation: the clusters *ps* and *mn*, and the ambisyllabicity of *x* [ks].

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3.1 Other clusters: *ps*, *mn*

The focus on *s impurum* clusters seems to be shortsighted, as these are a manifestation of a more general pattern involving any cluster that violates the SON SEQ constraint in one way or another, as exemplified by the clusters *ps* (stop-continuant, which is the reverse of canonical *s impurum*) and *mn* (nasal-nasal). These are infrequent, as word-initial occurrences are of Greek origin, (6)–(7). Notably, just as in the case of *s impurum* clusters, these clusters too are heterosyllabic when medial, (8)–(9).

- (7) *psō.ra* ‘itch, mange’
 (8) *Mnes.tēr*⁷ ‘a pantomime actor adored by Caligula’
 (9) *re.lāp.sus* ‘relapsed’
 (10) *am.nis* ‘a deep river,’ often figurative

Below, Tableau 3.1.1 illustrates the optimality of (6) *psō.ra* via the same constraint hierarchy featured earlier in Tableau 2.4.6, *stā.re*.

Tableau 3.1.1: PSŌRA ‘itch,’ ‘mange’

/psōra/, [psō.ra]

#ps is of Greek origin

/psōra/	MAX-I/O	DEP-I/O	SON SEQ
☞ (a) <i>psō.ra</i>			* [.ps]
(b) $\square$ p.sō.ra		*! > $\square$ <	
(c) p $\square$.sō.ra		*! > $\square$ <	
(d) sō.ra	*! < p >		

Above, given the input of word-initial /ps/, no alternative to output [ps] is possible in Latin. While speakers of other languages may delete or insert a segment as a “repair strategy,” these options are not permitted by the Latin constraint hierarchy, as we saw in Tableau 2.4.6, *stā.re*. The continued relevance of this tableau demonstrates that no additional theoretical apparatus is required for OT’s handling of the clusters *mn* and *ps*.

Below, Tableau 3.1.2 displays the optimality of (9) *am.nis* via the same constraint hierarchy previously seen in Tableau 2.4.5 for *res.tā.re*. As for the analogous *ps* cluster, the optimality of (8) *re.lāp.sus* is borne out by the same constraint hierarchy.

⁷ Considering the Classical Greek spelling Μνηστήρ, it is surprising that Gaffiot (1934:985) did not represent the first vowel of the Latinization *Mnestēr* as a long ē, as well.

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Tableau 3.2.2: AMNIS ‘deep river’

/amnis/, [am.nis]

/amnis/	SON SEQ	NO CODA
(a) a.mnis	*! [.mn]	* [s.]
☞ (b) am.nis		** [m.], [s.]

Since candidate (a)’s onset cluster fails to incur a sonority sequencing violation, it is eliminated right away, making (b) the winner, despite (b)’s two NO CODA violations.

In the following section, the proposed OT solution will be shown to provide consistent coverage for yet another case that the extrasyllabicity account does not address.

3.2 The ambisyllabicity of *x*

The robustness of the proposed analysis is evident in its effortless explanation of the phonological weight contributed by *x* [ks] to the preceding syllable, making it heavy (bimoraic). Classical grammarians refer to this phenomenon as “hidden weight.” For instance, consider the form *paradoxa* ‘paradoxes’. According to the basic conventions of word division, we would write *pa-ro-do-xa*, where each syllable has an onset. However, Allen and Greenough (1903) recommend analyzing this *x* as a coda, presumably because this letter does not occur word-initially in the native vocabulary. Despite the coda placement, however, they do acknowledge the ambisyllabic nature of *x* in practice: “**dix-it, sax-um**, etc. are preferred to **di-xit, sa-xum**; the pronunciation was probably **dic-sit, sac-sum**” (p. 4).

Phonologically, *x* [ks] is unquestionably ambisyllabic: [pa.ra.'dɒk.sa]. This means that, besides [s] serving as an onset, [k] is enlisted as coda to the preceding syllable. The ambisyllabicity of *x* is supported by a well-established, independently-motivated rule in Latin: stress falls on the antepenult when the penult is light (monomoraic). The coda [k] contributes additional weight to the penult, rendering it bimoraic, thus preventing antepenult (“paroxytonic”) stress and ensuring correct stress placement on the penult (“paroxytone”).

Conversely, if syllabification were *[pa.ra.do.ksa], with an onset [ks] cluster, stress would incorrectly fall on the antepenult, as the penult would be light and therefore unstressed. The fact that the vowel is short indicates that penult stress can only result from the weight provided by a coda, [k]. Thus, the surface form [pa.ra.'dɒk.sa], per the rule of stress placement, fortuitously coincides with the optimal output of the proposed constraint hierarchy, as demonstrated in the following tableau with a ranking that should by now be familiar, as this case parallels that of e.g. *res.tā.re*, where *s impurum* is medial.

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Tableau 3.2: PARADOXA ‘paradoxes’

/paradoksa/, [pa.ra.'dok.sa]

/paradoksa/	SON SEQ	ONSET	NO CODA
(a) pa.ra.do.ksa	*! [.ks]		
☞ (b) pa.ra.dok.sa			* [k.]
(c) pa.ra.doks.a		*! [.a]	** [ks.]

In comparison, the extrasyllabicity hypothesis for *s impurum* does not adequately account for the exceptional behavior of *x*, despite their identical metrical effects. Although word-initial *x* is extremely rare in Latin, examples such as *xenium* [kse.ni.um] ‘gift, esp. for/from a host’ and *Xerxēs* [kserk.sēs] ‘a King of Persia (485-465 BC)’ evidence an “inconsistency” in syllabification similar to that of *s impurum*.

That *x* [ks] is homosyllabic word-initially but ambisyllabic in medial position might lead one to consider the possibility of “*x impurum*” and “extrasyllabic *x*.” However, proposing such constructs would overlook the fact that these patterns are epiphenomenal manifestations of a more fundamental phonological phenomenon, as these tableaux reveal. Consequently, this extrasyllabicity approach must be discarded for failing to capitalize on generalizations and its lack of parsimony, thereby running afoul of Occam's Razor, which maintains that entities should not be multiplied beyond necessity

To summarize, we have architected a coherent account that successfully explains the *s impurum* data, the similarly contradictory syllabifications required of *ps* and *mn*, and the seemingly exceptional metrical effect of *x*. This OT analysis demonstrates remarkable robustness and parsimony, proving its effectiveness in addressing a range of phonological phenomena.

4. Debunking “extrasyllabicity”

The extrasyllabicity analysis will be outlined in Section 4.1, followed by a critique in 4.2.

4.1 Previous analysis (Cser 2012, 2016)

This section is to provide a review of the extrasyllabicity analysis presented by Cser (2012, 2016), which appears to represent the most recent treatment of the *s impurum* phenomenon. The extrasyllabicity analysis draws upon the concept of word-initial “extrasyllabic” [s] (Jakob 1992), which by definition is excluded from syllabification for the purposes of theoretical analysis.

A fundamental assumption of this analysis is that, since *s impura* cannot be syllabified as onsets word-medially, the same restriction must apply to all positions, including word-initially. In support of this claim, Cser invokes

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Occam’s Razor. However, the relevance of this heuristic to this case is undermined by a Fallacy of Composition (Section 4.2.2.1).

This heterosyllabic division of the *s impurum* cluster is obliged by a prohibition stating: “an extrasyllabic segment may not be adjacent to a nucleus” (Cser 2012:369). This compels the process of “leftward resyllabification,” a generative, rule-based transformation whereby the purportedly extrasyllabic [s], in {curly brackets} below, becomes a coda to the preceding syllable, (11)–(14).

(11) ho. + {s} tēs	→	hos. tēs	‘enemies’
(12) ve. {s} per	→	ves. per	‘evening’
(13) re. + {s} tat	→	res. tat	‘it remains’
(14) ob. + {s} tā. re	→	obs. tā. re	‘to oppose’

In effect, *s impurum* is “rescued” from its extrasyllabic condition by the preceding syllable.

Apart from *s impurum*, Cser (2012:366) notes that “there is no leftward syllabification of other stem-initial consonants,” (15)–(16).

(15) re. + frāc. ta	‘broken up’
(16) re. + tra. hō	‘I bring back’

In passing, the use of a designation as general as “other stem-initial consonants” is curious, as it absconds the critical role that sonority sequencing plays in distinguishing (11)–(14) from (15)–(16).

Of further relevance, stop + liquid clusters are heterosyllabic when there is an intervening morpheme boundary, (17)–(19).

(17) ab. + rup. ta	‘broken off’
(18) ob. + li. gō	‘I obligate, make liable’
(19) vi. det. # rem	‘sees the thing’

Assuming (17)–(19) to represent the default scenario, Cser notes that resyllabification exceptionally proceeds rightward to prevent the formation of a syllable without an onset following a closed syllable, (20)–(21).

(20) ab. + ī. re	→	a. bī. re	‘to go away’
(21) vi. det # il. le	→	vi. de. t#il. le	‘he sees’

Although these examples are presented in support of Latin extrasyllabicity, such an account does not fully integrate the fundamental basics of syllable structure, the role of sonority sequencing in syllabification, or morphological alignment to provide a coherent, unified explanation of these surface forms. The extrasyllabicity account is constrained by the framework of traditional generative phonology, which necessarily stipulates inviolable,

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explicitly ordered rules, ultimately overlooking higher-level generalizations that could shed valuable light on the forces shaping Latin's phonological grammar.

In contrast, an Optimality Theoretic approach harnesses these obvious and necessary generalizations to drive the analysis, addressing the issues of *s impurum*—and more—without resorting to extrasyllabicity and resyllabification. By leveraging violable, language-agnostic phonological constraints, this framework enables us to evaluate competing syllabifications without relying on ad hoc concepts and maneuvers.

4.2 Critique

The critique of the extrasyllabicity analysis will be divided into two primary areas of concern. First, we'll examine Cser's five theory-internal arguments supporting the extrasyllabic hypothesis, followed by a section dedicated to the more general reasoning and argumentation underpinning that analysis.

4.2.1 Critique of phonological arguments

In the following sections, we will review each of Cser's (2012:364–365) theory-internal arguments intended to support the extrasyllabicity hypothesis:

- 4.2.1.1 Atypical sonority sequencing
- 4.2.1.2 Inflectional *-s* suffix: *urbs*, *rēx*
- 4.2.1.3 Word-initial vs. -medial syllabification: *stāre* vs. *res.tā.re*
- 4.2.1.4 *S impurum* vs. obstruent-liquid clusters: *re.stā.re* vs. *re.frāc.ta*
- 4.2.1.5 *S*-less reduplication (rare)

4.2.1.1 Atypical sonority sequencing

Cser argues that *s impurum* clusters defy the Sonority Sequencing Principle, as the *s* is more sonorous than a stop, leading to a relative sonority peak between the stop to its right and a word/syllable boundary to the left, thus disrupting the consistent rise in sonority from the syllable edge towards the nucleus.

The observation that *s impurum* clusters do not conform to sonority sequencing alone is not a compelling argument for designating this [s] as “extrasyllabic” in Latin, just as it would be insufficient to argue the same for English. Furthermore, Cser does not specify a theoretical mechanism that would relate the sonority profile of certain phonotactic sequences to immunity from syllabification on the first round but not on a subsequent round, as supposed extrasyllabicity is only temporary in the case of *s impurum* medially, e.g. *res.tā.re* ‘to remain’, *ves.per* ‘evening’, *hos.tēs* ‘enemies’, *as.tra* ‘stars’. Rather than supporting the extrasyllabicity hypothesis, this phenomenon calls for a more comprehensive understanding of syllable structure and the principles by which sonority influences it.

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4.2.1.2 Inflectional -s suffix: *urbs*, *rēx*

Cser (2012:364) notes that word-final stop+s clusters only emerge when the inflectional suffix [-s] is added to stop-final stems, as in *urbs* and *rēx*. While this observation is accurate, attributing the pattern solely to extrasyllabicity is premature, as further evidence is needed to support this claim.

First, the very relevance of this observation must be questioned. stop+s clusters function as codas and are notably the reverse sequence of *s impurum*. It remains unclear how reversing this phonotactic sequence could provide information about the implications of such a rearrangement for extrasyllabicity. That is, even if we were to assume that word-final [s] is extrasyllabic, there is no apparent reason why *reversing* the order of the segments would preserve the extrasyllabicity of the original order. Intuitively, such an assumption does not seem to comport with established phonological principles and requires a more rigorous examination to support such claims.

It is likely relevant that [s]’s place of articulation is coronal. Cross-linguistically, a coronal place of articulation is common in coda (and therefore word-final) position, as observed in English words like *rind* (ending in [+coronal, +anterior]) and *range* (ending in [+coronal, -anterior])—in both cases, a long vowel (phonetically, a diphthong) precedes a **coronal** consonant. However, the same long vowel is prohibited when the final consonant is **labial**, **r[aj]mp* or velar-final **r[aj]nk* (Fudge 1969, Halle & Vergnaud 1980, Paradis & Prunet 1991). Observations such as these cross-linguistically have caught the attention of these researchers who acknowledge “the special status of coronals.” The phonologically uneventful suffixation of coronal [s] does not appear to justify the drastic exceptionality inherently associated with exemption from syllabification.

Extending Cser’s logic, then, one might wonder whether *all* syllable appendices should be considered extrasyllabic. By way of analogy, it would seem an ad hoc and opportunistic maneuver to label the coronal [d] of *rind* [rajnd] as extrasyllabic simply to account for the ill-formedness of English **r[aj]mp* and **r[aj]nk*.⁸ Relying on the contrived designation of “extrasyllabic,” such an approach fails to leverage the underlying principles that drive syllabification, such as sonority sequencing and markedness theory.

A final flaw in this argument is its reliance on circular reasoning. Cser engages in a “question-begging” fallacy by labeling inflectional [s] in *post-stop* clusters (*urbs*, *rēx*) as “extrasyllabic” without providing proof of its extrasyllabicity to begin with and by subsequently using this unsubstantiated claim to support his main argument—that the [s] in the reversed sequence must be extrasyllabic as well.

4.2.1.3 Word-initial vs. -medial syllabification: *stā.re* vs. *res.tā.re*

Cser contends that the organization of initial *s* + stop clusters, as in *stā.re*, parallel that of medial clusters, such as *hos.tēs*, where abundant metrical evidence indicates that the *s* belongs to the first syllable. This suggests that initial 's' + stop clusters, like 'st', are not syllable onsets.

⁸ The phonologically alien nature of **r[aj]mp* and **r[aj]nk* is evident in their defiance of conventional English orthography. Although awkward and unsightly, the approximations “righmp” and “righnk” serve as attempts to best represent these peculiar forms.

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Meanwhile, the OT analysis has demonstrated that the heterosyllabicity of the cluster in *res.tā.re* is due to the ready availability of an alternative syllabification that the environment provides, given the presence of a preceding vowel. No such option is available for *stā.re*, given that there is no preceding syllable available to house a coda [s]. In the absence of a superior alternative, onset position is optimal in this environment.

Cser's observation regarding forms like *stā.re* versus *res.tā.re* does not necessitate the conclusion that *s impurum* is extrasyllabic. By examining the interaction of constraints operating on phonemic inputs, we can better understand the underlying phonological mechanisms and develop more comprehensive explanations for syllabification patterns in Latin.

4.2.1.4 *S impurum* vs. obstruent-liquid clusters: *re.stā.re* vs. *re.frāc.ta*

Cser argues that the contrasting syllabification patterns of word-medial *s impurum* clusters, like *res.tā.re* (heterosyllabic and first syllable heavy), and obstruent-liquid clusters, like *re.frāc.ta* (homosyllabic and first syllable light), serve as evidence for extrasyllabicity, highlighting the distinction between the syllable divisions in these different contexts.

However, the difference in syllabification is not surprising, given that obstruent-liquid clusters comply with the sonority sequencing constraint, while *s impurum* clusters do not. Actually, the comparison between these distinct cluster types appears to be irrelevant to the syllabification of *s impurum*, and reaching the far-fetched conclusion of extrasyllabicity without considering sonority sequencing appears to be an extreme measure.

These syllabification patterns have been more effectively and insightfully explained through a hierarchy of ranked and violable constraints than by invoking extrasyllabicity. As demonstrated in Tableau 2.4.5, the constraint L-ALIGN (M, σ) prevents heterosyllabicity in **re+f.rāc.ta*, while heterosyllabicity in *res.tā.re* results from the ranking SON SEQ ≫ L-ALIGN (M, σ) ≫ NO CODA. Thus, the optimality of words like *res.tā.re* and the ill-formedness of **re.stā.re* can be attributed to the satisfaction of high-ranking sonority sequencing, which is achieved by committing violations of lower-ranking L-ALIGN and NO CODA.

The transactional aspect of the OT analysis is sophisticated, robust, and intuitively appealing, in contrast to the question-begging, stipulative nature of the extrasyllabicity analysis. Another strength of this OT analysis is its wide-ranging generality, as the hierarchical machinery selecting the optimal candidate never makes reference to [s], or to any specific segment. The level of generality at which OT operates ensures the analysis's applicability across different phonological contexts and contributes to its overall elegance.

4.2.1.5 *S-less reduplication (rare)*

Cser highlights three verbs with initial *s* + stop clusters that exhibit a unique reduplicated perfect form, unlike other verbs with complex onsets. These verbs are *stāre* ~ *stetī*, *spondere* ~ *spo.pondī*, and *scindere* ~ Pre-Classical *scicidī*. Notably, the reduplication process only copies the stop segment of the cluster, excluding the *s*, rendering this pattern exceptional.

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This pattern of *s*-less reduplication is indeed a fascinating, compelling observation, but attributing this rare phenomenon to the synchronic, extrasyllabic status of the initial [s] is questionable. It is significant to note that this unusual pattern is not productive in Latin, being deeply rooted in the language's history, which suggests that other factors may be at play. A more comprehensive explanation might involve exploring the historical development of these verbs and their reduplication patterns, rather than relying solely on the assumption of extrasyllabicity. Furthermore, the fact that reduplication only copies the stop portion of the onset cluster may offer insights into the language's phonological constraints or preferences at the time when these verbs underwent productive reduplication, potentially tracing back to a pre-Latin era.

4.2.2 Critique of argumentation

It is essential to address some concerns about the reasoning that undergirds the extrasyllabicity analysis. These can be summarized in six main points:

- 4.2.2.1 Fallacy of Composition
- 4.2.2.2 False Dichotomy
- 4.2.2.3 Manipulation of Terminology and Conceptual Ambiguity
- 4.2.2.4 Begging the Question (*Petitio principii*)
- 4.2.2.5 Confirmation Bias
- 4.2.2.6 Appeal to Mystery

4.2.2.1 Fallacy of Composition

The Fallacy of Composition occurs when one assumes that something true for a part must also be true for the whole. For example, claiming that all of California enjoys a Mediterranean climate simply because San Diego has one is an instance of this fallacy.

Cser's extrasyllabicity analysis similarly commits the Fallacy of Composition by assuming that since *s impurum* cannot be syllabified as an onset word-medially (e.g., in *res.tā.re*), the same restriction must hold word-initially. This assumption is unfounded and does not consider the distinct phonological forces acting on these distinct contexts.

Furthermore, invoking Occam's Razor to justify this overgeneralization is inadequate because it prioritizes simplicity at the expense of an accurate, comprehensive analysis. In doing so, Cser inadvertently undermines the very simplicity argument based on Occam's Razor by basing the extrasyllabicity analysis on a logically flawed foundation.

Fallacies of Composition reflect a tendency to overgeneralize in an attempt to simplify and explain a seemingly inexplicable complexity. This not only exposes the flaws in the underlying logic but, in this case, also highlights

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the importance of considering the broader context and potential differences between various phonological contexts within a word.

4.2.2.2 False Dichotomy

The extrasyllabicity analysis presents a false dichotomy by framing the issue as a choice between extrasyllabicity and no alternative explanations. This oversimplification disregards potential alternative explanations, such as [s] being syllabified, which could be explored through contemporary approaches or historical explanations.

False dichotomies can inadvertently mislead readers by limiting their perspectives and preventing them from considering other solutions. In this case, the either-extrasyllabicity-or-nothing reasoning not only excludes constraint-based approaches like Optimality Theory (introduced previously in 1993) but also hinders a comprehensive understanding of the problem.

4.2.2.3 Manipulation of Terminology and Conceptual Ambiguity

In the present context, the primary issue lies in the manipulation of terminology to suit a specific argumentative purpose. The term “extrasyllabic” exemplifies this problem, as it introduces conceptual ambiguity and conceals an inherent contradiction.

First, labeling a segment “extrasyllabic” (unsyllabified, unparsed) only for a positional prohibition to force its syllabification at a later stage can be compared to the workings of a Rube Goldberg machine—constructing complex machinery to achieve seemingly unremarkable results.

The appeal of an academic, Latin-Greek hybrid like “extrasyllabicity” obscures the underlying issue—the attempt to reconcile two incompatible choices. The manipulation of the “extrasyllabic” concept leads to a contradictory assertion: while [s] is present in the practical, physical realm of syllables that are spoken and heard, it remains unaccounted for in syllable structure when its omission facilitates theoretical analysis. This demonstrates a desire to “have it both ways,” which ultimately creates more confusion in understanding Latin phonology.⁹

In addition, the lack of a clear definition for “extrasyllabicity” and positive examples to illustrate and validate the phenomenon complicates the issue further. Without a clear, independent illustration of what extrasyllabicity looks like in Latin, its value becomes questionable. It appears to function as a stipulation among others rather than as a well-defined, useful phonological concept. To justify the use of extrasyllabicity, an objective exemplification and explanation of its phonological characteristics and role within the broader context of Latin syllabification are necessary.

Ultimately, extrasyllabicity seems to be an ad hoc construct manipulated to coerce the observed surface-level facts

⁹ This work's dismissal of Latin extrasyllabicity nonetheless supports the concept of extrametricality, essential for analyzing Latin stress placement (Prince and Smolensky 1993). Extrametricality remains a vital tool, as it addresses syllables or segments exceeding the typical count in a given meter.

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into compliance with a biased hypothesis.

4.2.2.4 Begging the Question (*Petitio principii*)

In Section 4.2.1.2, we saw that Cser offers inflectional word-final [s], as in *urbs* and *rēx*, as evidence for the extrasyllabicity of *s impurum*. This argument appears to rely on circular reasoning, the fallacy known as *petitio principii* or “begging the question.” Cser identifies the occurrence of word-final stop+s clusters as evidence for extrasyllabicity, but the extrasyllabic nature of these clusters themselves lack evidentiary support.

By attributing the unsubstantiated extrasyllabicity of word-final stop+s clusters to support the claim of *s impurum* extrasyllabicity, Cser inadvertently creates a self-referential loop. This renders the argument logically flawed and indicates the need for a more thorough examination of the phenomenon.

4.2.2.5 Confirmation Bias

Cser’s argument for the extrasyllabicity of *s impurum* appears to be influenced by confirmation bias. He seems to leverage any uncertainties or gaps in understanding to support the extrasyllabicity hypothesis, as evidenced by his use of allegedly extrasyllabic inflectional word-final [s] as proof for the extrasyllabicity of *s impurum*. This selective interpretation of patterns might be guided by strong preconceived beliefs, which can reinforce existing views on extrasyllabicity’s validity. Consequently, such an approach may hinder the pursuit of a more objective understanding of each apparent manifestation, as alternative explanations or perspectives are potentially overlooked.

4.2.2.6 Appeal to Mystery

The analysis invokes one enigmatic concept, extrasyllabicity, to explain another—namely, the seemingly inconsistent syllabification of words like *stā.re* and *spi.ce.re* with respect to their derived forms *res.tā.re* and *sus.pi.ce.re* and to underived forms like *hos.tēs*, *ves.per*, and *as.ra*.

An important question remains unanswered within the extrasyllabicity account: How and at what stage of derivation did *s* become extrasyllabic? It seems unlikely that *s* is marked “extrasyllabic” at the underlying level (hypothetically, /re+{s}tāre/), since its occurrence is predictable, always preceding a pre-vocalic stop. Furthermore, it is unclear—and hence the as-needed stipulations and ordered rules—why *s* would be marked impervious to syllabification at one stage of derivation yet would be subsequently compelled to syllabify as a coda due to a positional restriction prohibiting the adjacency of an extrasyllabic segment and a nucleus—a restriction for which there is no independent evidence.

Cser’s prohibition necessitating “leftward resyllabification” appears to be conveniently tailored to address the very segments that the extrasyllabicity hypothesis has already deemed exceptional by immunizing them against an earlier round of syllabification by decree. This raises concerns about the foundations of his analysis, as its reliance on a series of convenient, interdependent assumptions calls into question the validity and applicability of extrasyllabicity, given the lack of independent justification for these stipulations. The lingering mysteries

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surrounding these stages in the derivation process indicate that the extrasyllabicity account falls short in providing an intuitive and coherent account of *s impurum*'s seemingly exceptional behavior.

In contrast, OT offers a robust framework for understanding “unusual” phonological patterns. Through the lens of OT, the biased notion of [s] being exceptional or problematic when followed by a pre-vocalic stop consonant in the input becomes irrelevant and groundless. Syllabification is driven by principles of sonority sequencing, alignment, and other preferred configurations of syllable structure, rendering such biased metaphors, like “unclean” and “trapped” [s] (the topic of Section 6) baseless. Every surface realization is optimal for its context, and so-called *s impurum* is no exception, as we have seen.

5. Demystifying “non-neutral” *s impurum*

There is one particular phonological environment where poets appear to have avoided *s impurum*. The extrasyllabicity account fails to provide a satisfactory explanation for this puzzling gap in the data. In this section, we will explore this phenomenon and propose an account based on another type of alignment constraint that applies optionally in connected speech, at least in the recitation of poetry. This constraint may have been enforced prescriptively while falling out of style, potentially creating awkwardness when spoken aloud. This could explain the poets' avoidance of *s impurum* in this specific context.

5.1 “Neutral” vs. “non-neutral” position

In analyzing *s impurum*'s distribution in Latin poetry, Cser (2012) introduces a valuable distinction between “neutral” and “non-neutral” positions. A “neutral” position refers to instances where it is supposedly impossible to determine the syllable to which the [s] belongs or when the syllabification of *s* is irrelevant in terms of the poet's adherence to metrical requirements. For example, word-initial *s impurum* preceded by a bimoraic syllable is said to constitute a neutral position—that is, when it is preceded by a coda consonant (VC # sC), a long vowel ($\bar{V}$ # sC), or a glide (VG # sC). In these cases, the syllable to the cluster's left is already heavy, making the syllabification of *s* inconsequential to its weight.

Another neutral position, purportedly, is the so-called “trapped” [s] of C[s]C sequences, whose syllabification Cser (2016) claims to be indeterminate, e.g. *dexter* [dekster] ‘right’. Cser capitalizes on this indeterminacy to advocate for the term “extrasyllabic.” However, see Section 6, “C[s]C: Setting ‘trapped’ [s] free.”

In contrast, *s impurum* occupies a “non-neutral” position whenever it is preceded by a *word-final short vowel*. This is the one phonological environment where there is no possibility of indeterminacy. The poet must take a stand on the syllabification of non-neutral *s impurum*: either it is syllabified as a coda to the preceding short vowel, in which case the syllable scans as heavy (**H**, bimoraic), as exemplified by Catullus in (22), or it is syllabified with its accompanying stop as an onset cluster, leaving the preceding syllable light (**L**, monomoraic), as demonstrated by Lucretius, further below in (1).

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(22) Non-neutral *s impurum* parsed as a coda

Catullus, *Carmina* 64.186

Nulla fugae ratiō nulla spēs omnia mūta

(nul-la fu)₁-(gae ra-ti)₂-(ō nul)₃-(**la s**-pēs)₄ (om-ni-a)₅ (mū-ta)₆
(H L L)₁ (H L L)₂ (H H)₃ (**H H**)₄ (H L L)₅ (H L)₆

Metrical analysis of Catullus's above verse confirms *s impurum*'s coda status and the weight it accordingly confers to its syllable. In this dactylic hexameter, the fourth foot houses the juncture of *nulla* 'no' and *spēs* 'hope', which scans as the expected spondee [las.pēs], i.e., a sequence of two heavy syllables, rather than as the iamb [la.spēs] (light – heavy), which would ensure alignment but would unfortunately break the meter. Scansion thus requires a coda analysis of non-neutral *s impurum* as in (22).

As a preliminary to the corresponding OT analysis offered in the next section, let us for now recognize the following phonological generalization: Although the heterosyllabicity of the cluster in *nul.la#s.pēs* entails misalignment, it strategically preempts a sonority sequencing issue—just as in the case of the word-medial cluster in *re.stā.re*.

Cser's data make clear that three poets take a stand opposite from Catullus in this matter: Horace, Propertius, and in (23), Lucretius.

(23) Non-neutral *s impurum* parsed as an onset

Lucretius, *De rerum natura* 4.475

Unde sciat quid sit scīre et nescīre vicissim

(un-de sci)₁-(at quid)₂ (sit scī)₃(r<e> et nes)₄-(cī-re vi)₅-(cis-sim)₆
(H L L)₁ (H H)₂ (H H)₃ (H H)₄ (H L L)₅ (H H)₆

The second and third syllable of a dactyl must always be light; thus, in the first foot of (23), the second syllable *de* is light and must therefore not have a coda. Consequently, the *s impurum* serves as an onset to the third syllable. Phonologically, Lucretius' verses demonstrate the inclination to prioritize the alignment of lexical and syllable boundaries over the circumvention of sonority sequencing issues.¹⁰ This generalization will be further explored and formalized in the next section.

¹⁰ This brings to mind German's preference for Alignment over the Onset constraint, contrasting with French, which displays the opposite preference, favoring Onset over Alignment. The consequent "blurring" of word boundaries in French, which could pose challenges to non-native speakers' listening comprehension, may have a language-processing parallel influencing syllabification decisions made by the Latin poets in their own language.

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5.2 An Optimality-Theoretic understanding of non-neutral *s impurum*

Cser (2012:367) notes that among the many poets surveyed, “Catullus applies leftward resyllabification in 5 out of 6 cases.” Since OT has no way to execute resyllabification, Catullus’s preference can be stated non-derivationally as: “*S impurum* is best syllabified with the preceding syllable *even at word boundaries*.” This preference satisfies the SON SEQ constraint by compromising alignment and is schematized in Tableau 5.2.1, which repurposes the constraint hierarchy from Tableau 2.4.5, *res.tā.re*.

Tableau 5.2.1: Catullus, NULLA SPĒS ‘no hope’

/nulla # spēs/, [nul.las.pēs]

/nulla spēs/	SON SEQ	L-ALIGN (M, σ)	NO CODA
(a) nul.la. # spēs	*! [.sp]		*[s.]
☞ (b) nul.la # s.pēs		* [+s.p] ¹¹	** [s.], [s.]

The above tableau reveals that Catullus prioritizes unmarked sonority sequencing over alignment at the word boundary. The same syllabification pattern that is standard word-medially, e.g. *res.tā.re*, also obtains at the word boundary in Catullus, as demonstrated by optimal *nul.la#s.pēs*, hence the renewed relevance of this constraint hierarchy.

Now, to address the alternate syllabification preferred by Horace, Propertius, and Lucretius, (23) and again in Tableau 5.2.2, we invoke another alignment constraint, L-ALIGN (LXWD, σ). Pertaining to the well-formedness of connected speech, it requires that the left edge of the lexical word be in alignment with that of the syllable. As we see in Tableau 5.2.2, in poets such as Lucretius, this lexical alignment constraint outranks SON SEQ.

Tableau 5.2.2: Lucretius, UNDE SCIAT ‘whence he would know’

/unde skiat/, [un.de. # ski.at]

/unde skiat/	L-ALIGN (LXWD, σ)	SON SEQ	L-ALIGN (M, σ)
☞ (a) un.de.#ski.at		* [.sk]	
(b) un.de#s.ki.at	*! [#s.]		*[+s.]

Tableau 5.2.2 demonstrates a preference for showcasing lexical alignment with the syllable boundary over preventing a sonority sequencing violation.

¹¹ A word boundary (#) implies a morpheme boundary (+).

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The syllabification of non-neutral *s impurum* differs among poets and can thus be explained in terms of constraint satisfaction. In Lucretius, Horace, and Propertius, it can be understood as the fallout of L-ALIGN (LXWD, σ) » SON SEQ. In the opposing camp, Catullus has no use for the lexical alignment constraint. By disregarding it, Catullus treats word-initial non-neutral *s impurum* as if the environment were word-medial, syllabifying it as a coda, as previously demonstrated in Tableau 5.2.1.

5.3 The significance of non-neutral *s impurum* & the poets' avoidance

Cser's comprehensive analysis highlights an intriguing pattern among the poets: an avoidance of *s impurum* in non-neutral position. The incidence of *s impurum* in this position is very low: of the 10,217 occurrences of all such words in the Cser's poetic corpus only 49 (or 0.48%) are found in this context (Cser 2012:367). The general pattern of avoidance is more pronounced in Golden and Silver Age poets, as Cser aptly notes: "Vergil is strongly averse to this configuration" (p. 367), and emphasizing the deliberateness of this avoidance, he asserts that the poets of both literary periods "are at pains to avoid having to take sides" (p. 369).

The motivation behind this avoidance is particularly intriguing, considering that the cluster seems to be systematically absent in the very position where poets could have expressed their intended syllabification due to the position's "non-neutrality."

When the cluster does appear, though rarely so, there is a lack of clearcut regularity in terms of which syllabification to expect. This issue raises further questions about the extrasyllabicity account, particularly regarding the reluctance to regularize the purported anomaly (i.e. syllabify the extrasyllabic) at the word boundary. That is, it seems peculiar that most poets, except for Catullus, cautiously prefer the syllabification that preserves alignment and thus—on Cser's account—maintains *s impurum*'s exceptional, extrasyllabic status, yielding the strange parse [un.de.{s}ki.at], where bracketed *s impurum* resists prosodic integration.

The explanation to be proposed here is speculative, and I invite fellow scholars to engage in a debate that may either support or challenge its premises. It arises in part from the observation that people often do not apply prescriptive grammar rules consistently, for whatever reason, which reveals their unnatural, alien nature in the context of everyday language use.

Such rules frequently exhibit a bias rooted in social exclusivity, portraying colloquial speech as inferior to the language found in Great Literature, whose perceived superiority presumably stems from its organization and tidiness on the page, the impression of pomp it gives, and its adherence to tradition. Prescriptive rules often derive from sources outside the realm of everyday language use, shaped by the personal preferences and prejudices of self-designated "language experts" and imposed on the general public.

In the context of *s impurum*, when considering which syllabification might have been enforced prescriptively, it is more likely to be (23) *un.de.#ski.at* (Lucretius), which prioritizes lexical alignment over unmarked sonority sequencing, than (22) *nul.la#s.pēs* (Catullus), displaying the reverse preference. This judgment is based on the type

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of prescriptivist rule that favors speakers who show an awareness of a form's presence in writing. The syllabification exemplified by Lucretius' (23) makes explicit reference to the left edge of the word, a feature more readily apparent in written language. As a result, prescriptive rules like this implicitly grants privilege or special status to the conventions of written language, reinforcing its supremacy over the natural patterns found in spoken language. The works of Plautus and Terence suggest a prescriptive ban on non-neutral *s impurum*, which could have been challenging to avoid. This prohibition is described as "words ending in short vowel [...] are not permitted before word [sic] beginning with *s impure*" (Cser 2016:138, citing Hoenigswald 1949), emphasizing the awkwardness or contrived nature of this phonological intolerance. Along these lines, Cser (2016:30) gives an example of this very sort of bias in the prescriptive pronunciation of /Vns/ and /Vnf/ that was current "in educated circles," where [n] was pronounced, presumably due to its presence in the *spelling* of these words.

Against the backdrop of poets avoiding or being reluctant to allow *s impurum* to exhibit "non-neutral" syllabification, Catullus stands out by uniquely adopting a "pro-sonority" stance, as demonstrated in Tableau 5.1.1 with *nul.la#s.pēs*. In doing so, he ignores the L-ALIGN (LXWD, σ) constraint, which he might have perceived as an unwelcome prescriptivist imposition, something along the lines of "Stop running your words together!" His blatant disregard for this constraint might have been a deliberate departure from the prescriptive tendencies or stylistic preferences of his time.

Catullus's idiosyncratic treatment of non-neutral *s impurum*, which he chose to versify as a coda—i.e., as if it were word-medial—seems consistent with his bold poetic persona. Renowned for his unconventional and profane style, his poetry dominated a range of registers—from anger to playfulness—to express scorn and personal insults, often characterized by strong emotion, obscenity, and audacity.

Catullus's divergence from other poets appears to be his way of implicitly expressing his views on linguistic "correctness" in Latin, essentially advocating for the acceptability of "blending" (i.e. misaligning) words in speech. By applying a word-medial phonological pattern (*ves.per*, *res.tā.re*) across word boundaries in verse (*nul.la#s.pēs*), Catullus might be reflecting the colloquial pronunciation of his time, possibly challenging traditionalist views held by "purists" who value conservatism and social exclusivity.

While it may be a stretch to expect "user-centered design thinking" from poets—let alone the Great Poets of Ancient Rome—the general avoidance of *s impurum* in non-neutral position might be attributed to attempts to minimize the cognitive burden that this superficially ambiguous syllable structure imposes on the audience. English features nearly ambiguous, "oronymic" examples, such as the popular children's chant, "I scream / You scream / We all scream / For ice cream!" and the phrase "Pat's car must be new," which, when spoken, can be easily confused with "Pat's scar must be new," the distinction being the aspirated [k] in "car." These amusing examples take advantage of the complex (and cross-linguistically marked) sonority sequencing of the [ksk] cluster to generate ambiguity. These amusing examples capitalize on the complex (and cross-linguistically marked) sonority profile of the [ksk] cluster to generate ambiguity. The double meaning results from the unexpected

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association of [s] with its other neighbor, creating a sense of misdirection and confusion. In particular, it is crucial to recognize the significance of the auditory aspect in the perception and enjoyment of ancient poetry, especially considering that literacy rates in Ancient Rome were relatively low. While reading poetry was common among the educated, listening was the primary way most people experienced and appreciated it.

Whether consumed auditorily or visually, the mental effort required for parsing non-neutral *s impurum* can be summarized in a systematic disambiguation rule such as the following: When the poetic meter requires a heavy syllable, the cluster must be heterosyllabic; otherwise, when the meter requires a light syllable, the cluster must be syllabified homosyllabically in onset position. Processing this unusual arrangement in the course of the comprehension process demands cognitive resources from readers or listeners, potentially increasing the likelihood of misreading, mishearing, or needing to pause to reanalyze. Of course, the option to pause for reflection is a luxury afforded only to readers, as recited poetry of the time did not offer the convenience for the listener/reader to pause or rewind. By avoiding *s impurum* in non-neutral position, poets in effect spared their listeners and readers the cognitive task of having to disambiguate the structure by reconciling syllabification with the meter, ultimately streamlining the experience and reducing potential confusion.

The use of non-neutral *s impurum* might have been avoided by poets in an attempt to navigate various sociolinguistic challenges. In addition to potential language-processing issues, this structure could have faced criticism from those who viewed themselves as guardians of the linguistic “clarity,” “purity,” and “logic” of Latin. Furthermore, time-honored linguistic traditions, such as syllabifying *s impurum* with its respective morpheme, might have created phonotactic issues, which could have been perceived as awkward or pedantic when spoken aloud. These factors highlight the tension between prescriptive constraints and natural speech patterns that poets had to balance.

The picture presented here is a no-win situation. The poet faces the dilemma of either compromising connected-speech naturalness by complying with a strict (i.e., high-ranking) alignment constraint, or facing potential criticism for failing to yield to prescriptive pressures by ignoring it.

Poets who syllabified non-neutral *s impurum* as a coda faced an additional risk of generating ambiguity beyond the structural. They themselves could have been perceived as either intentionally dismissing the prescriptive alignment rule for stylistic reasons, or as being ignorant or careless in apparently overlooking the lexical boundary. It is not surprising, then, that most poets chose to avoid non-neutral *s impurum*, opting for a safer, less controversial, and less conspicuous approach that minimized attention to the cluster and its impact on prosody.

Examining *s impurum* within these contexts sheds light on the intriguing relationship between phonological constraints, stylistic choices, and possible prescriptive biases in Latin poetry, as well as how poets like Catullus challenged established conventions.

6. C[s]C: Setting “trapped” [s] free

Our final area of inquiry delves into one subpattern of *s impurum* whose syllabification has lacked consensus—sequences consisting of [s] between two consonants (C[s]C), as in *extrā* [ekstrā] ‘outside’, *dexter* [dekster] ‘right’, where the intervening [s] is described as “trapped” (Cser 2016:60).

Table 2, below, showcases examples of C[s]C, with the first three cases notably featuring a morpheme boundary preceding the *s impurum*. The table displays optimal outputs selected by the proposed constraint hierarchy alongside the syllabifications resulting from Cser’s extrasyllabicity proposal. This comparison aims to highlight the differences between the two approaches and emphasize the robustness of the proposed constraint hierarchy in resolving these particular cases.

TABLE 2: C[s]C

	Input	OT output [s] = coda	Per Cser (2016) [s] = extrasyllabic	Gloss
<i>absque</i>	/abs+k ^w e/	[aps.k ^w e]	[ap.{s}.k ^w e]	‘except for’
<i>abscēdō</i>	/abs+kēdō/	[aps.kēdō]	[ap.{s}.kē.dō]	‘I walk away’
<i>abscondō</i>	/abs+kondō/	[aps.kon.dō]	[ap.{s}.kon.dō]	‘I conceal’
<i>constō</i>	/kon+stō/	[kōs.tō] ¹²	[kon.{s}.tō]	‘I stand firm’
<i>dexter</i>	/dekster/	[deks.ter]	[dek.{s}.ter]	‘right’ (masc. sg.)
<i>extrā</i>	/ekstrā/	[eks.trā]	[ek.{s}.trā]	‘outside’ (adv.)

Analysis of these sequences presents a challenge because the syllable preceding the “trapped” [s] is already heavy, due to its existing coda. This factor makes it impossible to infer, through metrical analysis, which syllable the [s] belongs to. Consequently, this would appear to be a case of *s impurum* in a “neutral” position, as its syllable affiliation is presumed to be unknowable through metrical analysis.

Cser (2016) leverages this presumed conundrum to endorse Jacobs’ (1992) use of “extrasyllabic” over Lehmann’s (2005) use of “post-coda” and “pre-onset.” Cser argues that referring to *s impurum* as “extrasyllabic” eliminates the need to decide whether the “trapped” [s] is pre-initial or post-coda “on an ad-hoc basis” (Cser 2016:60, fn 96).

¹² Word-medially, coda /n/ only surfaces before homorganic stops: e.g [nt], [mb]. Before S and F, the nasal fails to surface, though the preceding vowel is compensatorily lengthened and nasalized.

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Tableau 6.1: CONSUMPTA ‘consumed’

/kon+sumpta/, [kõ.sũmp.ta]

/kon+sumpta/	SON SEQ	L-ALIGN (M, σ)	No CODA
☞ (a) kõ.sũmp.ta			*** [n.], [mp.]
(b) kõ.sũm.pta	*! [.pt]		** [n.], [m]

Tableau 6.2: ABSQUE ‘except for’

/abs+k^we/, [aps.k^we]

/abs+k ^w e/	SON SEQ	L-ALIGN (M, σ)	No CODA
☞ (a) aps.k ^w e			** [ps.]
(b) ap.sk ^w e	*! [.sk]	* [p.s+]	* [p.]

It is surprising and puzzling that a coda analysis of “trapped” [s] would be characterized as an “ad hoc” decision, given that (1) there is no independent motivation in Latin to syllabify this [s] as an onset, and (2) the coda parse simultaneously forestalls sonority sequencing violations *and* ensures proper alignment. This win-win situation is visually represented in Tableaux 6.1–6.2 by the adjacent blank cells to the right of each candidate (a).

7. Summary

By reexamining these phenomena Latin phonology through the lens of Optimality Theory, we can appreciate that each realization of /s/ is optimal in its particular environment, thus challenging the implication that *s impurum*, as characterized by Cser’s extrasyllabicity solution, is exceptional. The extrasyllabicity analysis reveals the potential pitfalls of an inherited bias in linguistic research, as it may have led to a misunderstanding of the phonological principles at play. Ultimately, this exploration of *s impurum* serves as a reminder that impartiality and open-mindedness are essential in uncovering the true nature of language.

Cser appeals to Occam’s Razor in support of his extrasyllabicity analysis; however, this heuristic much more robustly endorses the proposed OT analysis. First, through the lens of OT, we have demonstrated that the very notion of extrasyllabicity is unnecessary to account for the observed patterns. Secondly, in any OT approach, constraints themselves are fundamental not only in capturing generalizations but also in driving the analysis. As a result, we have avoided ad hoc constructs such as resyllabification and extrasyllabicity, which have effectively obscured the orchestration of principles fundamental to the phonological analysis of Latin.

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Unlike generative, derivational approaches that primarily provide technical restatements and schematizations of the data, the proposed OT analysis seeks to offer deeper insight by leveraging these constraints to illuminate Latin phonology as a whole, emphasizing parsimony, simplicity, and generalizability over mere technical descriptions of data sets.

In this light, we can reconsider the allegedly exceptional nature of *s impurum* and instead reevaluate the data within the broader context of Latin phonological patterns. By doing so, we can attribute the behavior of *s impurum* to the natural consequences of the constraint hierarchy represented as the following dominance chain.

(24) Constraint ranking

MAX-I/O » DEP-I/O » SON SEQ » ONSET » L-ALIGN (M, σ) » NO CODA

These constraints, ranked as such, not only address the syllabification of *s impurum* but also account for broader phonological patterns characteristic of Latin pronunciation. For example, we have observed that this hierarchy generates the correct outcomes for obstruent-plus-liquid clusters, whether they are homosyllabic (e.g. *re.frāc.ta*) or heterosyllabic (e.g. *ab.rup.ta*).

The same constraint hierarchy successfully explains the seemingly inconsistent syllabification of clusters such as *ps* and *mn* and the ambisyllabicity of onset *x*—without requiring additional theoretical stipulations or machinery. Thus, this analysis offers a concise yet comprehensive account of various phenomena, demonstrating the efficacy of a constraint-based approach in addressing linguistic issues that might initially appear enigmatic and unrelated when examined through a traditional derivational lens.

We have also seen that OT can successfully elucidate the behavior of *s impurum* in non-neutral position, where the extrasyllabicity account falls short. By introducing a high-ranking constraint on word-syllable alignment, which might have been applied prescriptively rather than colloquially, we can explain the tendency of word-initial *s impurum* to optionally syllabify as an onset, as in the works of Lucretius, Horace, and Propertius, but not in prone-to-insult-and-obscenity Catullus.

This hypothesis also offers insights into the poets' psychology, as it suggests they were averse to committing to a particular approach to parsing non-neutral *s impurum* one way or the other, or they were unwilling to allow potential ambiguity or stylistic awkwardness in their work, contributing to the general avoidance of *s impurum* in this position.

As for the so-called “neutral” C[s]C position, we have observed that the consistent and coherent nature of the Latin constraint hierarchy dictates that the purportedly “trapped” [s] must be parsed as a coda. This conclusion is further supported by an independently established, fundamental orthographic rule for dividing Latin consonant sequences (e.g., *con.sūmp.ta*), which reinforces the internal consistency and validity of the system.

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We have also seen that the argumentation underpinning the extrasyllabicity analysis appears to be influenced by a convergence of logical fallacies. For example, the Fallacy of Composition can be observed in the overgeneralization of specific cases. The Appeal to Mystery solves one enigma by invoking another. Additionally, the argument engages in question-begging, presents readers with a false dichotomy, and manipulates the meaning of “extrasyllabic” in an attempt to reconcile two incompatible positions simultaneously.

In contrast, the OT analysis avoids these logical pitfalls through its steadfast reliance on inevitable generalizations to drive the analysis, rather than by making assumptions as abstract as that of extrasyllabicity and resyllabification. By evaluating candidates against ranked and violable constraints, the OT approach offers a more rigorous, principled, and coherent understanding of syllabification in Latin, grounded in widely recognized phonological principles.

Problems once “solved” by generative approaches can now be reinterpreted through the lens of Optimality Theory. This is an intellectual exercise through which we arrive at an overarching, coherent constraint hierarchy, reminiscent of Gestalt psychology, which emphasizes the discernment of large-scale patterns and configurations, beyond their individual components. Again, the proposed analysis never specifies [s], or any segment. In opposition, the traditional generative approach restricts the researcher’s attention to addressing “problematic components,” which has apparently led to overlooking the epiphenomenality shared by *s impurum*, other clusters like *ps* and *mn*, and ambisyllabic *x*—an issue that can be attributed to narrow focus.

Through the emergent Gestalt (“shape”) of this analysis, we can appreciate Latin phonology as an organic and coherent whole, with its wide-ranging systematicity revealed through our deepening understanding of its underlying integrity.

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